Design and Analysis of a 2-Bit Magnitude Comparator using Cadence Virtuoso

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***Abstract*—This project presents the design, implementation, and analysis of a 2-bit magnitude comparator using Cadence Virtuoso. The comparator determines the relative magnitude between two 2-bit binary numbers, producing outputs indicating greater-than, equal-to, and less-than conditions. Initially, the logical behavior of the circuit was verified using Logisim by constructing and simulating the digital schematic and comparing it with the expected truth table. The transistor-level schematic, layout, and post-layout simulations were then performed in Cadence Virtuoso to analyze physical characteristics such as propagation delay, average power, and verification checks including DRC (Design Rule Check) and LVS (Layout Versus Schematic). The results demonstrate functional accuracy, correct logical behavior, and optimized power consumption for the implemented design.**

***Index Terms*—2-bit magnitude comparator, Logisim, Cadence Virtuoso, DRC, LVS, post-layout simulation, power analysis.**

I. Introduction

The magnitude comparator is a fundamental combinational circuit used in arithmetic and logical operations to compare two binary numbers. It forms the basis of several digital systems such as sorting networks, arithmetic logic units (ALUs), and control circuits. The purpose of this project is to design and analyze a 2-bit magnitude comparator at both the logic and transistor levels using Logisim and Cadence Virtuoso, respectively.

This project aims to:

- Design and verify the truth table of a 2-bit magnitude comparator using Logisim.
- Develop the transistor-level schematic in Cadence Virtuoso.
- Generate the physical layout and ensure design correctness through DRC and LVS checks.
- Analyze performance metrics such as power and propagation delay through simulation.

II. Theory

A 2-bit magnitude comparator compares two binary numbers, $A = A_1A_0$ and $B = B_1B_0$, and produces three outputs:

- $A > B$
- $A = B$
- $A < B$

Each bit is compared starting from the most significant bit (MSB). The logical equations for the comparator outputs are derived from the truth table as follows:

$$A > B = A_1\overline{B_1} + (A_1 \odot B_1)A_0\overline{B_0}$$

$$A = B = (A_1 \odot B_1)(A_0 \odot B_0)$$

$$A < B = \overline{A_1}B_1 + (A_1 \odot B_1)\overline{A_0}B_0$$

where $\odot$ denotes the XNOR operation.

A. Truth Table

The truth table for the 2-bit magnitude comparator is given in Table I.

TABLE I: Truth Table of 2-Bit Magnitude Comparator

A1	A0	B1	B0	A>B	A=B	A<B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

III. Methodology

A. Logisim Implementation

The logical circuit of the 2-bit magnitude comparator was implemented in **Logisim Evolution** to validate its functionality. The design used basic logic gates (AND, OR, NOT) to realize the Boolean equations derived above. Two sets of 2-bit binary inputs (A_1, A_0, B_1, B_0) were provided, and the three output conditions ($A > B$, $A = B$, $A < B$) were observed directly on the output pins of the simulator.

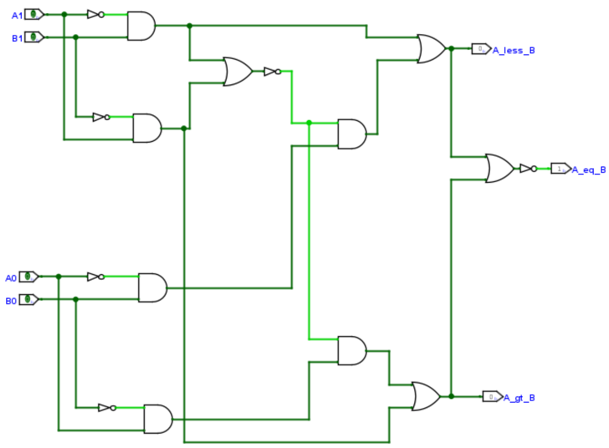


Fig. 1: Implementation of 2-Bit Magnitude Comparator in Logisim.

The output responses were tested against all possible input combinations to confirm that the circuit follows the expected truth table behavior.

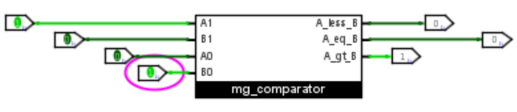


Fig. 2: Truth Table Verification of 2-Bit Magnitude Comparator in Logisim (for input A1=1, A0=0, B1=0, B0=1, showing that A is greater than B).

B. Cadence Virtuoso Implementation

After verifying logical functionality, the circuit was implemented at the transistor level in Cadence Virtuoso. The schematic was designed using NMOS and PMOS transistors in CMOS logic configuration.

1) Schematic Design: The schematic was created in the Virtuoso Schematic Editor. Each logical gate was designed using complementary CMOS logic. Power (VDD) and ground (GND) were connected appropriately. The design was checked for connection integrity before simulation.

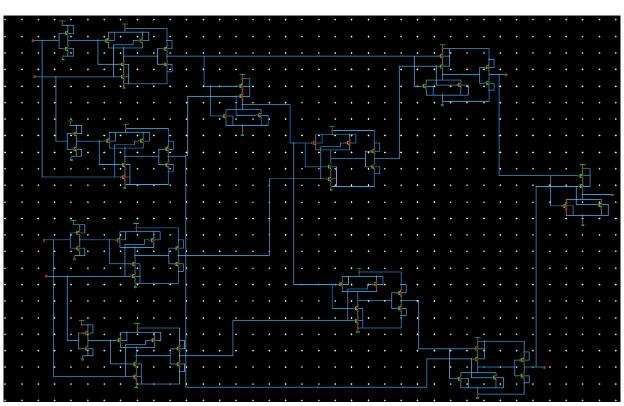


Fig. 3: Schematic Diagram of 2-Bit Magnitude Comparator in Cadence Virtuoso.

2) Layout Design: The layout of the 2-bit magnitude comparator was implemented in the Virtuoso Layout Editor following standard CMOS design rules. A new layout cell, `Mag_comp`, was created as the foundation of the design.

The display grid and snapping were configured to ensure precise placement of components: minor spacing of 0.01 µm, major spacing of 0.1 µm, and X/Y snap spacing of 0.005 µm. Design Rule Driven (DRD) notifications were enabled to maintain compliance with all layout constraints.

The logic gates used in the design were constructed using CMOS transistors. AND, OR, NOR gates, and inverters were placed modularly according to the comparator's Boolean equations, ensuring symmetric transistor placement to reduce parasitic effects.

Metal1, Metal2, Metal3 and Metal4 layers were employed for routing signals between gates. Inputs A_1, A_0, B_1, B_0 were routed from the left, while outputs A_less_B, A_eq_B, A_gt_B were routed to the right. Vias were inserted at Poly-to-Metal1 junctions for proper connectivity. Power (VDD) and ground (GND) rails were defined on Metal2 with 0.5 µm width and connected to Metal1 via standard via stacks.

Pins were assigned for all inputs and outputs with proper layer and direction settings. Additional Metal1 rectangles were added to meet minimum area requirements. The layout was finalized after passing DRC and LVS checks, confirming that the physical implementation matched the schematic and adhered to all design rules.

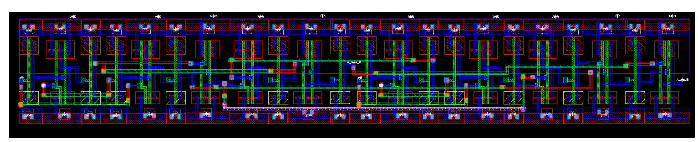


Fig. 4: Complete Layout of 2-Bit Magnitude Comparator.

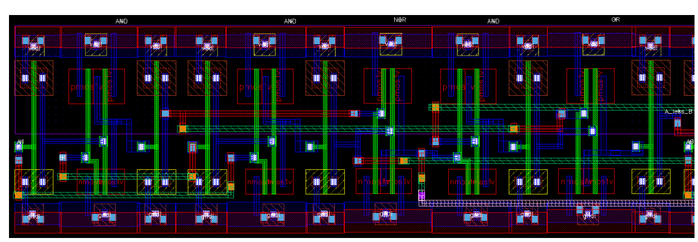


Fig. 5: First Half of the Layout.

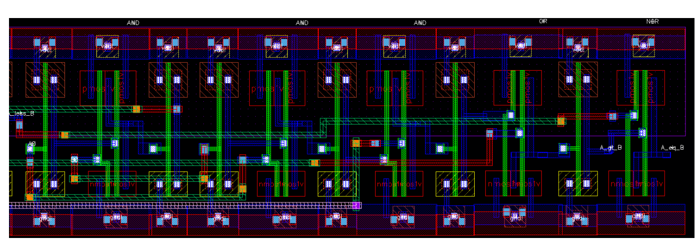


Fig. 6: Second Half of the Layout.

3) DRC and LVS Verification: Design Rule Check (DRC) was performed to ensure that the layout satisfies all physical design rules. Layout Versus Schematic (LVS) verification confirmed that the layout matches the schematic connectivity.

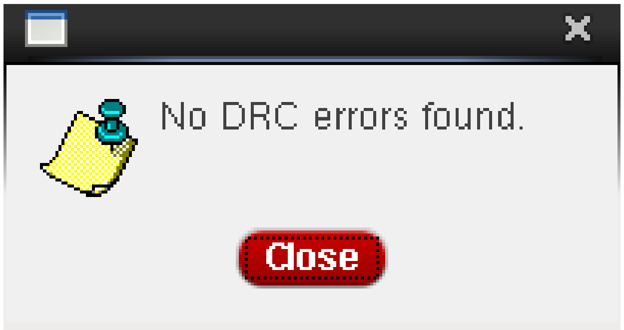


Fig. 7: DRC Check Result.

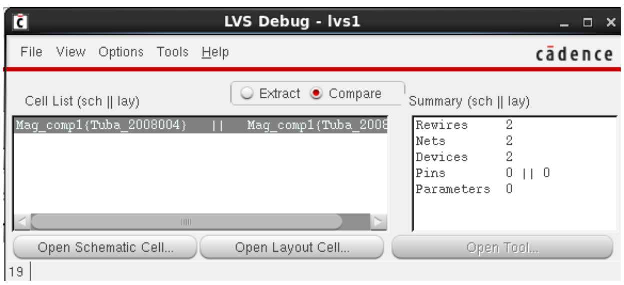


Fig. 8: LVS Error Debug Window.

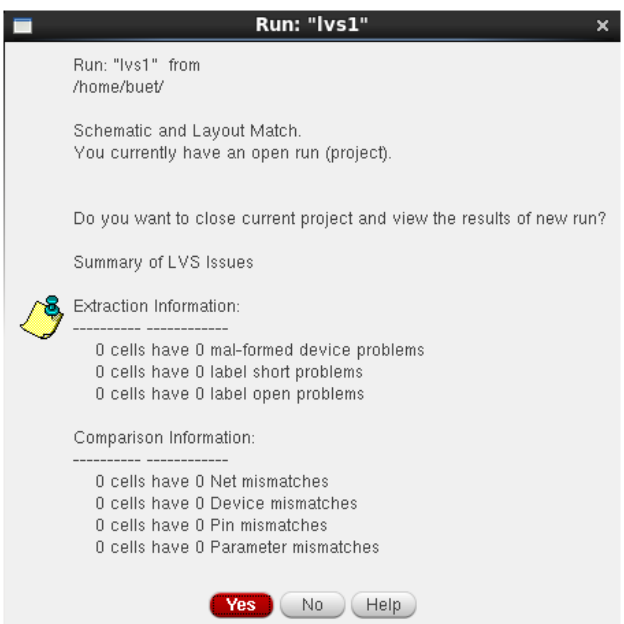


Fig. 9: LVS Verification Result.

4) Parasitic Extraction and Post-Layout Simulation: An extracted view of the circuit was generated to include parasitic resistances and capacitances. This lays the ground for post-layout simulation.

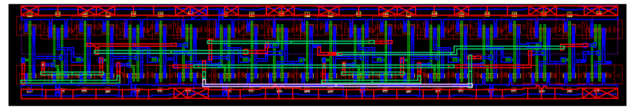


Fig. 10: Extracted View of the 2-Bit Magnitude Comparator.

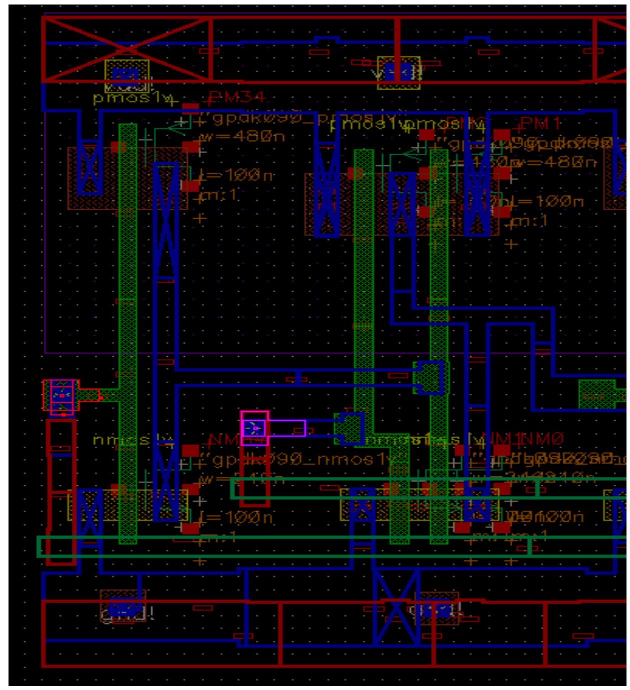


Fig. 11: Input A1 and B1 Signal.

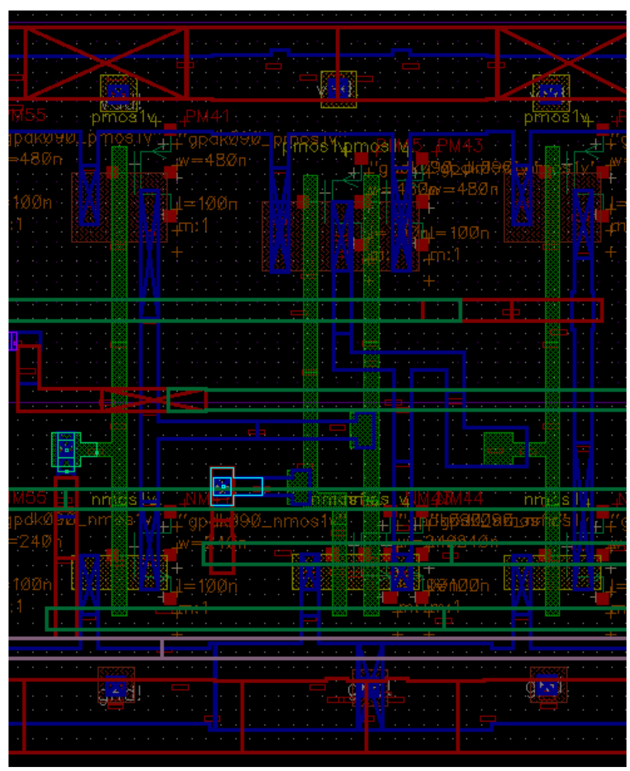


Fig. 12: Input A0 and B0 Signal.

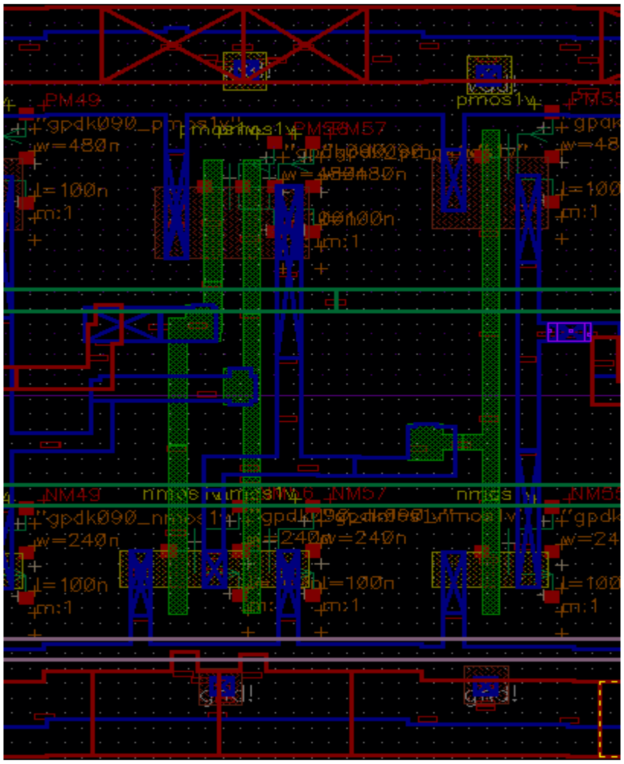


Fig. 13: Output A_less_B Signal.

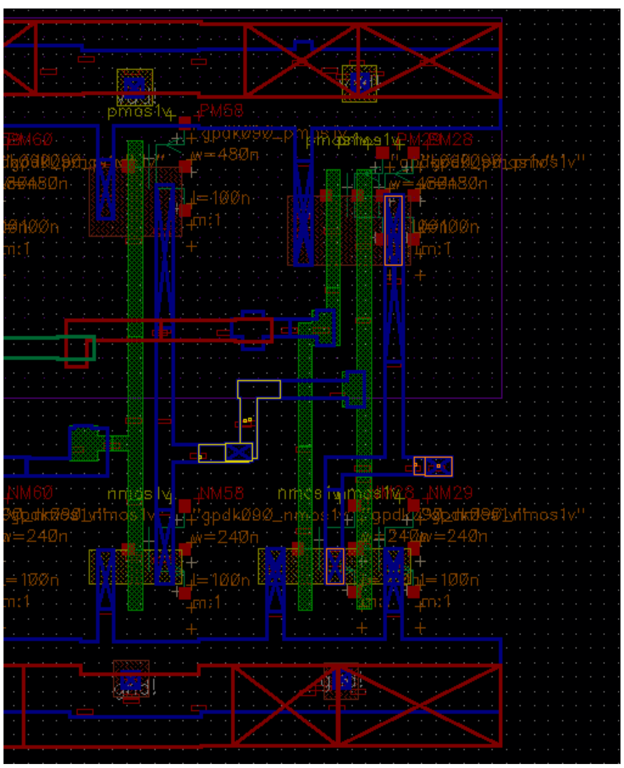


Fig. 14: Output A_gt_B and A_eq_B Simulation/Signal.

IV. Analysis

The performance of the 2-bit magnitude comparator was analyzed through schematic-level and post-layout (extracted) simulations using Cadence Virtuoso with the Spectre simulator. Both simulations were used to evaluate rise and fall characteristics, power consumption, and waveform integrity of the three outputs: **A>B**, **A<B**, and **A=B**.

A. Waveform Comparison

The transient simulation waveforms of the schematic and extracted netlists were compared to verify functional and timing consistency. Both simulations produced identical logical results in accordance with the expected truth table. However, the extracted waveforms exhibited slightly slower transitions due to parasitic resistances and capacitances introduced by interconnects in the layout. These parasitics contributed to minor delay and power variations, as discussed below.

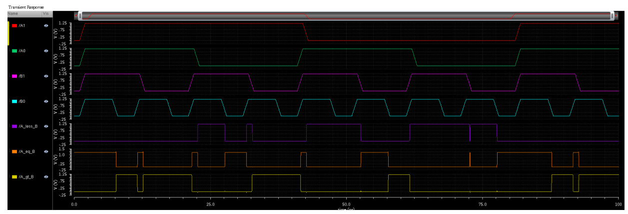


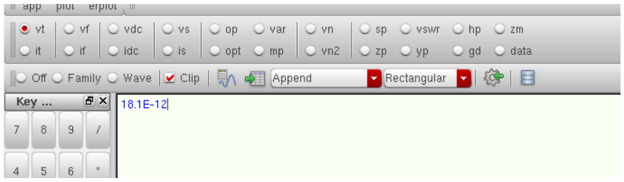
Fig. 15: Output waveforms of the 2-bit comparator.

B. Rise and Fall Time Analysis

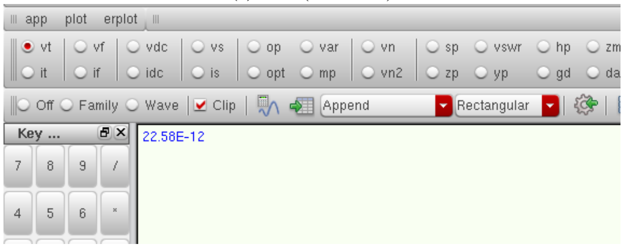
The rise and fall transition characteristics for each output were measured using the waveform calculator. Table II summarizes the observed rise/fall behavior derived from the transient waveforms. Although propagation delay could not be accurately computed due to the multiple-input dependency of the comparator, relative transition speeds between the schematic and extracted levels were used for comparison. The extracted circuit consistently showed slower transitions, confirming the parasitic impact on dynamic behavior.

TABLE II: Rise/Fall Time Comparison Between Schematic and Extracted Simulations

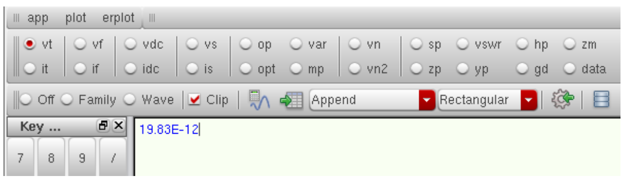
Output	**Schematic (s)**	**Extracted (s)**
A=B	18.1×10^{-12}	22.58×10^{-12}
A<B	19.83×10^{-12}	32.98×10^{-12}
A>B	22.07×10^{-12}	22.69×10^{-12}



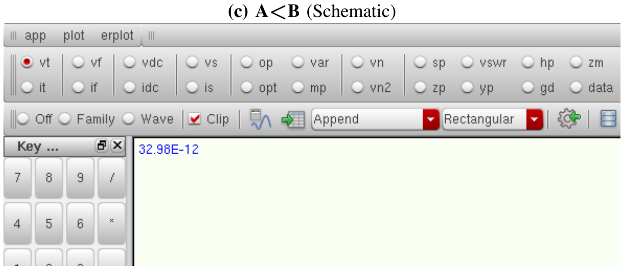
(a) A=B (Schematic)



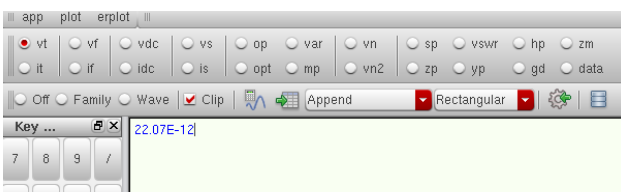
(b) A=B (Extracted)



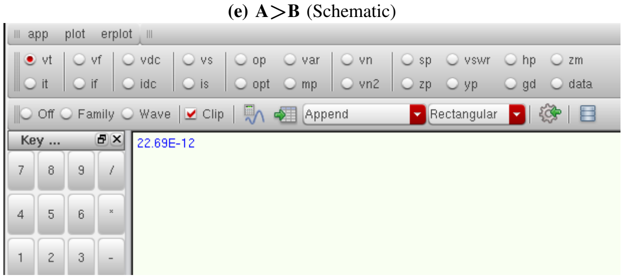
(c) **A<B** (Schematic)



(d) **A<B** (Extracted)



(e) **A>B** (Schematic)



(f) **A>B** (Extracted)

Fig. 16: Rise/Fall Time Waveform Comparison Between Schematic and Extracted Circuits

C. Power Analysis

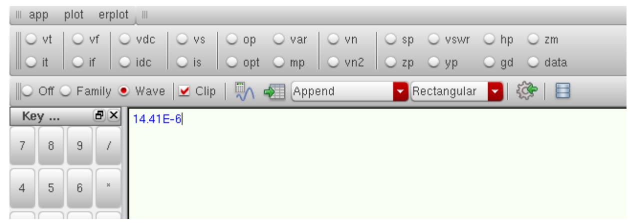
The average power of the comparator was computed using the waveform calculator in ADE L, considering the supply voltage and current through VDD. The average power was computed using the *average()* function in the calculator as:

$$P_{avg} = -\text{average}(V_{DD}(t) \times I_{VDD}(t))$$

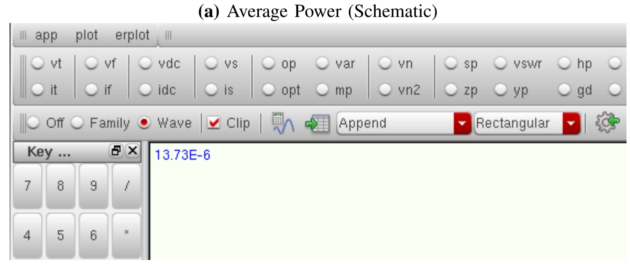
The measured average power from the schematic and extracted simulations is summarized in Table III.

TABLE III: Average Power Comparison Between Schematic and Extracted Circuits

Parameter	**Schematic (W)**	**Extracted (W)**
Average Power	14.41×10^{-6}	13.73×10^{-6}



(a) Average Power (Schematic)



(b) Average Power (Extracted)

Fig. 17: Average Power Comparison Between Schematic and Extracted Circuits

The schematic-level simulation shows an average power consumption of 14.41×10^{-6} W, while the post-layout extracted simulation reports 13.73×10^{-6} W. The slight reduction in post-layout power is attributed to the redistribution of switching currents and parasitic interactions that alter effective load capacitances during transitions. Although parasitics generally tend to increase power, specific interconnect effects or marginally slower signal transitions can result in a modest overall reduction.

Among the three outputs, $A < B$ exhibited the largest increase in delay, rising from 19.83 ps in the schematic to 32.98 ps in the post-layout simulation, indicating greater sensitivity to interconnect parasitics along that path. In contrast, $A > B$ showed minimal delay variation (from 22.07 ps to 22.69 ps), suggesting a more balanced and shorter routing configuration. The output $A = B$ exhibited a moderate delay increase from 18.1 ps to 22.58 ps, consistent with the influence of moderate parasitic loading.

These observations confirm that logical correctness was preserved across both schematic and extracted simulations, while timing and power characteristics were affected by realistic parasitic effects. Such variations are typical in post-layout CMOS designs, where wire capacitances and resistive paths influence the circuit's dynamic response.

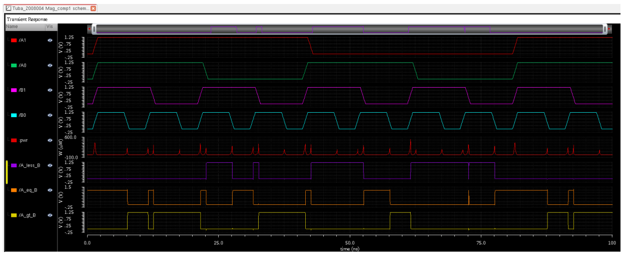


Fig. 18: Power Waveform of the 2-bit comparator.

V. Symbol Design

The symbol for the 2-bit magnitude comparator was created from the verified schematic in the Virtuoso Symbol Editor. Input and output pins (*A1, A0, B1, B0,* $A > B$, $A < B$, *A*=*B*) were defined to match the schematic connections.

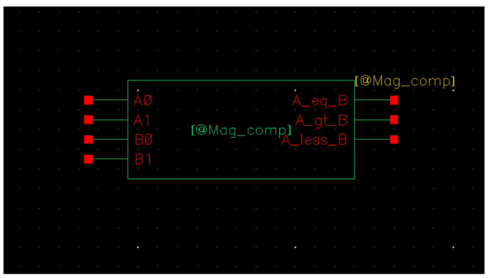


Fig. 19: Symbol of 2-Bit Magnitude Comparator in Cadence Virtuoso.

VI. Results and Discussion

The post-layout simulation results validate that the 2-bit magnitude comparator functions correctly and maintains full consistency between schematic and layout implementations. Functional verification confirmed that the output logic levels correspond precisely to the truth table for all input combinations. The DRC and LVS checks reported zero errors, ensuring structural and electrical integrity of the final layout.

Quantitative measurements indicate that parasitic effects introduced during layout moderately affect circuit performance. The propagation delays of the outputs `A_eq_B`, `A_less_B`, and `A_gt_B` were observed as 22.58 ps, 32.98 ps, and 22.69 ps respectively, compared to 18.1 ps, 19.83 ps, and 22.07 ps in the schematic simulation. This increase in delay for certain outputs is expected due to interconnect capacitance and resistance in the extracted layout.

The post-layout average power was measured at 13.73 μW, compared to 14.41 μW from schematic simulation. The measured reduction in average power in the extracted view is small (approximately $\approx 4.72\%$) and can result from differences in how parasitics redistribute switching currents, subtle changes in transition slopes, or differences in the averaging interval or stimulus alignment between runs. Regardless, both schematic and extracted values are on the same order of magnitude and within acceptable CMOS power limits.

Overall, the post-layout simulations closely match the schematic results, with only minor variations in timing and power. These results demonstrate that the comparator design is robust: logical functionality is preserved and the physical implementation exhibits realistic performance characteristics suitable for VLSI integration.

VII. Conclusion

The design and verification of the 2-bit magnitude comparator were successfully completed using Cadence Virtuoso. The schematic and layout were verified through DRC and LVS with zero errors, confirming full physical and logical consistency. Post-layout simulations demonstrated reliable operation, acceptable propagation delays, and comparable average power (schematic: 14.41 μW; post-layout: 13.73 μW).

The modest differences between schematic and extracted simulations are attributable to post-layout parasitic effects and potential measurement/averaging differences; these are typical in CMOS design flows. The overall results confirm that the implemented comparator achieves correct functionality, energy-efficient performance, and design integrity appropriate for integration into larger digital systems.

References

[1] VLSI Laboratory Manual, Department of Electronics and Telecommunication Engineering, Chittagong University of Engineering and Technology (CUET), Bangladesh, 2025.